

Part A

①

- ① State & explain the first and second law of Thermodynamics with illustrations.

A:- When a system undergoes a thermodynamic cycle then the net heat supplied to the system from the surrounding is equal to net work done by the system and on the surroundings.

$$\oint dQ = \oint dW.$$

Second law:-

Clausius statement:-

It is impossible for a self acting machine working in a cyclic process unaided by any external agency, to convey heat from a body at a lower temperature to a body at a higher temp.

Kelvin-Planck statement

It is impossible to construct an engine which while operating in a cycle produces no other effect except to extract heat from a single reservoir and do equivalent amount of work.

2. A spherical balloon holds 5kg of air at 200 kPa / 450K. If the air pressure inside is proportional to the square of the balloon diameter. determine the work done when the volume doubles due to heating.

Sol

$$m = 5 \text{ kg}, P = 200 \text{ kPa} / 450 \text{ K}.$$

$$P_1 V_1 = m R T_1$$

$$200 \times 10^3 \times V_1 = 5 \times 8314 \times 450$$

$$V_1 = 93.53 \text{ m}^3.$$

$$\therefore V_2 = 2V_1 = 187.065 \text{ m}^3.$$

$$\begin{aligned} \therefore \text{work done} &= P \Delta V \\ &= 200 \times 10^3 \times (93.53) \\ &= 18706. \text{ KJ.} \end{aligned}$$

2: A heat engine, a heat pump and a refrigerator receive 500 kJ of heat each. But they reject 250 kJ, 600 kJ, & 700 kJ of heat respectively. Determine i) the efficiency of the heat engine ii) the COP of the heat pump and the refrigerator.

Sol:

(i) efficiency of heat engine $\rightarrow$
 $Q_1 = 500 \text{ kJ}$
 $Q_2 = 250 \text{ kJ}$

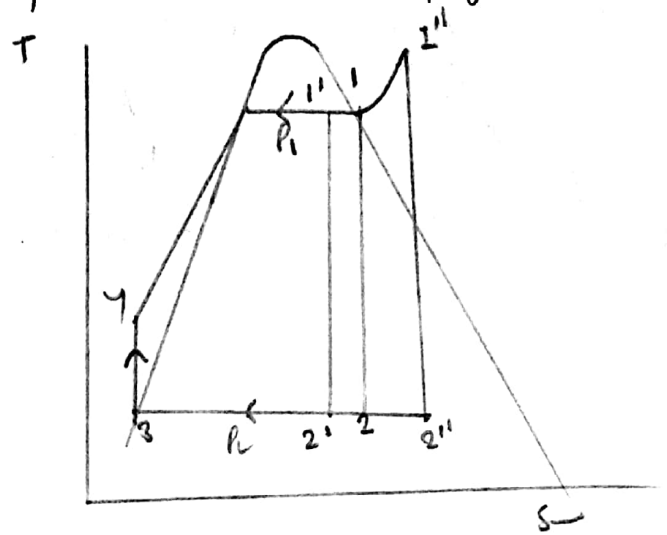
$$\begin{aligned}\eta_{\text{heat engine}} &= \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1} \\ &= \frac{500 - 250}{500} \\ &= \frac{250}{500} = \frac{1}{2} \\ &= 50\%\end{aligned}$$

$$\begin{aligned}\text{(ii) COP of heat pump} &= \frac{T_1}{T_1 - T_2} = \frac{Q_1}{Q_1 - Q_2} \\ &= \frac{500}{500 - 250} = \frac{500}{250} = 2\end{aligned}$$

$$\begin{aligned}\text{(iii) COP}_{\text{ref}} &= \frac{Q_2}{Q_1 - Q_2} = \frac{700}{500 - 700} = \frac{700}{-200} \\ &= -3.5\end{aligned}$$



4. Briefly explain steam generation process with the help of T-S diagram.



5. Explain Rankine cycle with the help of T-S diagram.

Process 1-2:-

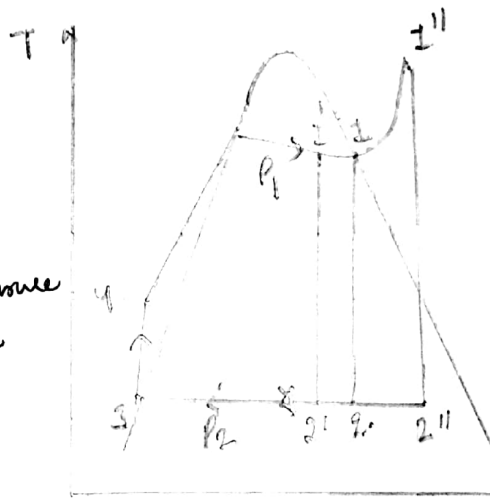
Reversible adiabatic expansion in the turbine.

Process 2-3:- constant-pressure transfer of heat in the Condenser.

Process 3-4: reversible

adiabatic pumping process in the feed pump

Process 4-1: constant-pressure transfer of heat in the boiler



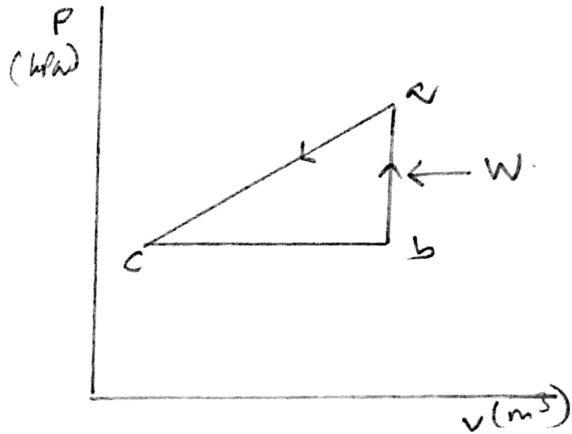
6. What are the desirable characteristics of a working fluid in a vapour cycle?

A:-

- 1- The fluid should have high critical temp.
- 2- The saturation pressure at the temperature of heat rejection should be above atmospheric pressure
- 3- The specific heat of the liquid should be small.
- 4- Saturated vapour line of the T-S diagram should be steep.

Part-B

- I: (a) A gas enclosed in a cylinder - piston assembly undergoes processes ab, bc & ca as shown. Given $P_a = 200 \text{ kPa}$ & $P_b = 100 \text{ kPa}$. Calculate the work done in the cycle. Also indicate the nature of the work.



- (b) An angry person slapped a man in a face which caused the temperature of the affected area to rise by 1.8 K . Assume the slapping hand has a mass of 1.25 kg and about 0.100 kg of the tissue of the face and the hand was absorbed by the unfortunate incident. As a forensic engineer, determine the velocity of the hand just before the impact with the face. Take the specific heat of the affected tissue to be 3.20 kJ/kgK . Take suitable assumptions for the evaluation.

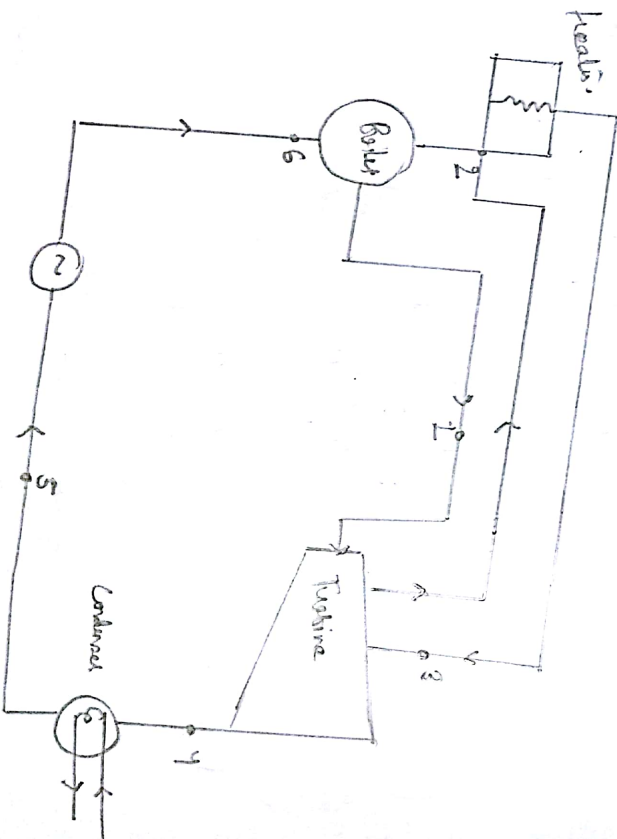
- A



Carnot engine (cycle) gives the highest thermal efficiency which is given by $\frac{T_1 - T_2}{T_1}$. To approach this cycle in an actual engine it is necessary that source of heat must be supplied at constant temperature T_1 & rejected at T_2 . This can be achieved by using vapours in the wet field but not in the superheated.

(b)

Low-Atchard Positive reheat improves the cycle efficiency by about 1% for a 65/15 bar cycle. A second reheat will give a much gain unless the initial cost involved would be too high as to permit use of two stage reheat except in case of very high initial throttle conditions. The cost of reheat equipment consisting of boiler piping and controls may be 5% to 10% more than that of the main boiler and this additional expenditure is justified only if gain in thermal efficiency is sufficient to provide a return of the investment.



Reheat cycle.

$$\frac{18}{100} = \frac{1}{100} = 1$$

$$38.258 = h_1 - h_2 = 1$$

$$h_1 - h_2 = 147.64$$

$$h_3 = h_4 @ 0.05 \text{ bar}$$

$$h_2 = 2028.36$$

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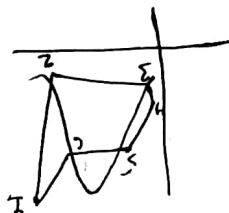
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31st July - 7th August

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Sol. (c)

$$1 \text{ ton of refrigeration} = 3.5 \text{ kW (given)}$$

$$= 3.5 \frac{\text{kJ}}{\text{s}}$$

$$= \frac{3.5 \text{ kJ} \times 60}{\text{min}}$$

$$= 210 \text{ kJ/min.}$$

$$\text{Cooling req.} = 30 \text{ ton}$$

$$= 30 \times 210$$

$$= 6300 \text{ kJ/min.}$$

$$T_1 = 303 \text{ K}$$

$$T_2 = 263 \text{ K.}$$

$$\text{COP} = \frac{T_2}{T_1 - T_2} = \frac{263}{303 - 263} = 6.57.$$

$$\text{Actual COP} = 0.5 \times 6.57$$

$$= 3.2875.$$

$$\text{COP} = \frac{\text{Cooling req.}}{\text{actual work need}}$$

$$W = \frac{6300}{3.2875}$$

$$= 1916.34981 \text{ kJ/min.} = 114980.4$$

$$= 31939 \text{ kJ/sec.}$$

$$\therefore \text{Cost} = \frac{31.939 \times 5.25 \times 720 \text{ h.} \times 114980.4}{3600}$$

$$= 120729.42 \text{ rupees}$$

A

$$W_{a-c} = \int_a^c p dv = -p_a v_a \ln \frac{v_c}{v_a}$$

$$= 202 \times 10^3 \times 0.001 \ln \frac{v_c}{v_a}$$

$\therefore W_{a-b} = 0 \because$ const volume.

$W_{b-c} =$ work done at const pressure.